



Restructuring Development & Implementation of ISRU in NASA's Artemis Program

Presenter's Name
Luxembourg Space Resources Week

May 20, 2025

Evolution of STMD Lunar ISRU Development & Mission Plans

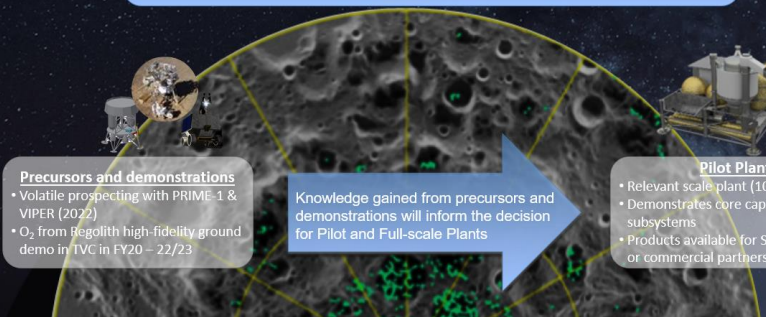


From the start, the Driving Strategy for ISRU in STMD was to try to Break the 'Chicken & Egg' cycle by Demonstrating ISRU Systems and Product at Relevant Mission Scale to eliminate the risk for Full-Scale Commercial Operations

Lunar ISRU Strategy: Leader/Follower

STMD has a **leader/follower** path defined for lunar ISRU

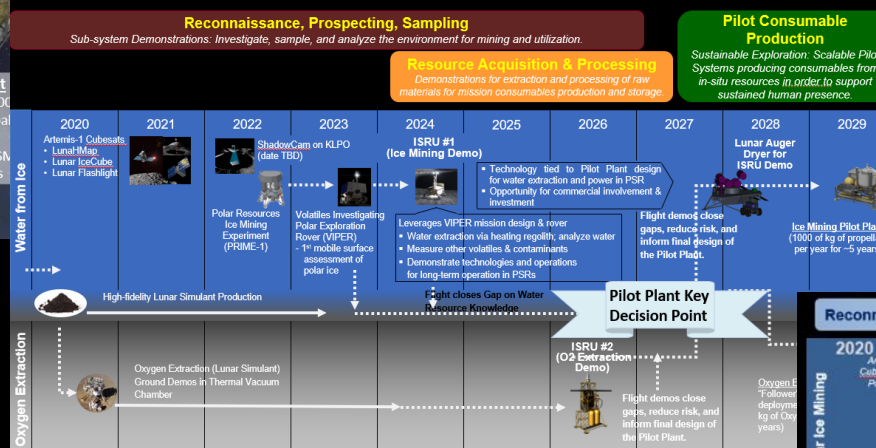
- **Ice mining** (Leader) – Potential to provide LO_2/LH_2 propulsion, crew consumables, and water for radiation protection
- **O_2 from regolith** (follower) – Provide oxidizer and crew consumables



The emphasis on Enabling Industry to be Successful has grown over time

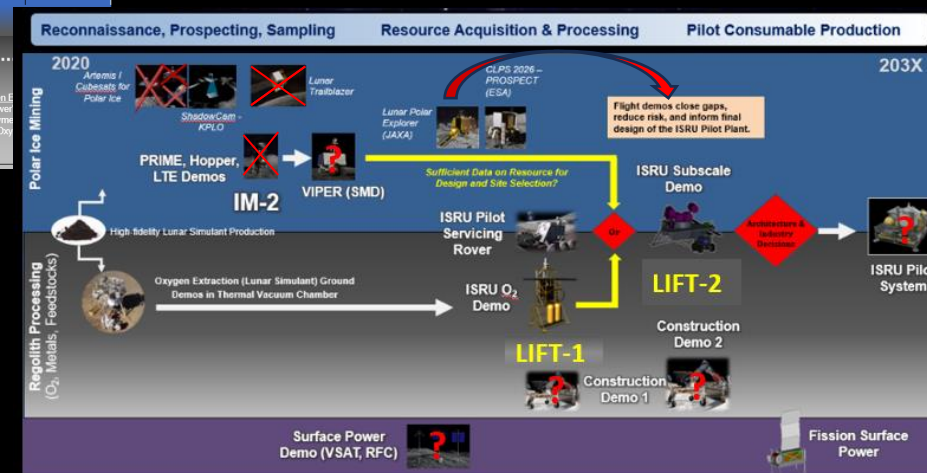
Lunar ISRU Development and Demonstration Timeline

OMB Chart 9/2020



However

Mission failures, cancellations, delays in Artemis missions, and budget cuts have forced a reevaluation of priorities, development approach, and flight implementation



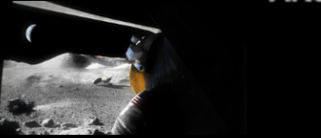
Evolution of Moon to Mars Objective Decomposition for ISRU



Artemis Architecture Segments

HUMAN LUNAR RETURN

Initial capabilities, systems, and operations necessary to reestablish human presence and initial utilization (e.g., science) on and around the Moon.



Locate Resources

Demonstrate ISRU

Map Resource Reserves

Utilize ISRU

FOUNDATIONAL EXPLORATION

Expansion of lunar capabilities, systems, and operations supporting complex orbital and surface missions to conduct utilization (e.g., science) and Mars-forward precursor missions.

SUSTAINED LUNAR EVOLUTION

Enabling capabilities, systems, and operations to support regional and global utilization (e.g., science), economic opportunity, and a steady cadence of human presence on and around the Moon.

HUMANS TO MARS

Initial capabilities, systems, and operations necessary to establish human presence and initial utilization (e.g., science) on Mars and continued exploration.

Demonstrate & Utilize ISRU



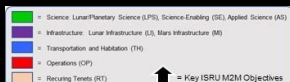
NASA Moon to Mars (M2M) Blueprint Objectives – ISRU Relevant

Resource Assessment	
AS-3 ^M	Characterize accessible lunar and Martian resources, gather scientific research data, and analyze potential reserves to satisfy science and technology objectives and enable In-Situ Resource Utilization (ISRU) on successive missions.
OP-3 ^M	Characterize accessible resources, gather scientific research data, and analyze potential reserves to satisfy science and technology objectives and enable use of resources on successive missions.
LPS-3 ^M	Reveal inner solar system volatile origin and delivery processes by determining the age, origin, distribution, abundance, composition, transport, and sequestration of lunar and martian volatiles.
TH-7 ^M	Develop systems for crew to explore, operate, and live on the martian surface to address key questions with respect to science and resources.
SE-3 ^M	Develop the capability to retrieve core samples of frozen volatiles from permanently shadowed regions on the Moon and volatile-bearing sites on Mars and to deliver them in pristine states to modern curation facilities on Earth.

Responsible ISRU	
SE-7 ^M	Preserve and protect representative features of special interest, including lunar permanently shadowed regions and the radio quiet far side as well as Martian recurring slope lineae, to enable future high-priority science investigations.
OP-12 ^M	Establish procedures and systems that will minimize the disturbance to the local environment, maximize the resources available to future explorers, and allow for reuse/recycling of material transported from Earth (and from the lunar surface in the case of Mars) to be used during exploration.
LI-8 ^L	Develop environmental monitoring, situational awareness, and early warning capabilities to support a resilient, continuous human/robotic lunar presence.
RT-6	Responsible Use: conduct all activities for the exploration and use of outer space for peaceful purposes consistent with international obligations, and principles for responsible behavior in space.

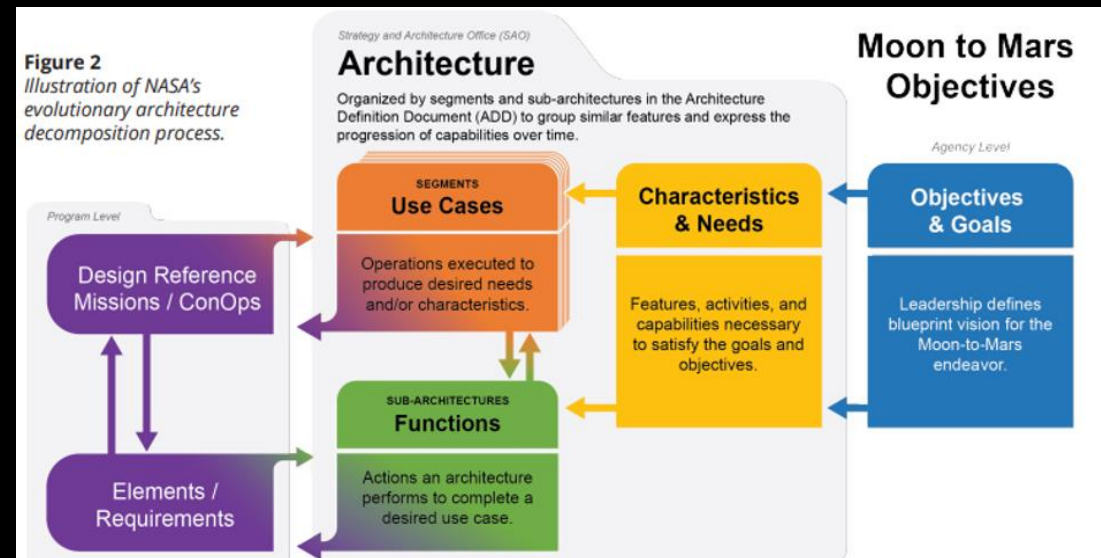
ISRU and Usage	
LI-7 ^L	Demonstrate industrial scale ISRU capabilities in support of continuous human lunar presence and a robust lunar economy.
LI-8 ^L	Demonstrate technologies supporting cislunar orbital/surface depots, construction and manufacturing maximizing the use of in-situ resources, and support systems needed for continuous human/robotic presence.
MU-4 ^M	Demonstrate Mars ISRU capabilities to support an initial human Mars exploration campaign.
OP-13 ^M	Demonstrate the capability to use commodities produced from planetary surface or in-space resources to reduce the mass required to be transported from Earth.
TH-3 ^L	Develop system(s) to allow crew to explore, operate, and live on the lunar surface and in lunar orbit with scalability to continuous presence, conducting scientific and industrial utilization as well as Mars analog activities.

These are the Recurring Tenets; themes common across all blueprint objectives.	
RT-1	International Collaboration: partner with international community to achieve common goals and objectives.
RT-2	Industry Collaboration: partner with U.S. industry to achieve common goals and objectives.
RT-3	Crew Return: return crews safely to Earth while mitigating adverse impacts to crew health.
RT-4	Crew Time: maximize crew time available for science and engineering activities within planned mission durations.
RT-5	Maintainability and Reuse: when practical, design systems for maintainability, reuse, and/or recycling to support the long-term sustainability of operations and increase Earth independence.
RT-6	Responsible Use: conduct all activities for the exploration and use of outer space for peaceful purposes consistent with international obligations, and principles for responsible behavior in space.
RT-7	Interoperability: enable interoperability and commonality (technical, operations and process standards) among systems, elements, and crews throughout the campaign.
RT-8	Leverage Low Earth Orbit: leverage infrastructure in Low Earth Orbit to support M2M activities.
RT-9	Commerce and Space Development: foster the expansion of the economic sphere beyond Earth orbit to support U.S. industry and innovation.



Artemis Architecture Definition Document, Rev. B

Figure 2
Illustration of NASA's
evolutionary architecture
decomposition process.



Functions	Use Cases	Characteristics/Needs	Objectives
Transport a moderate amount of cargo (1000s of kg) from Earth to south pole region sites on the lunar surface	FN-T-202 L		
Provide power for deployed external surface utilization payload(s) and/or equipment for long durations (months to years+)	FN-P-402 L		
Enable crew lunar surface extravehicular activity in sunlit areas and non-PSRs	FN-M-101 L		
Unload a moderate amount of cargo (1000s of kg) on the lunar surface	FN-M-402 L		
Reposition a moderate amount of cargo (1000s of kg) at the south pole region on the lunar surface	FN-M-503 L		
Provide communications and data exchange between the lunar surface and Earth	FN-C-101 L		
Conduct ISRU utilization payload and/or equipment operations on the lunar surface	FN-C-104 L		
Conduct resource identification utilization payload and/or equipment operations on the lunar surface	FN-I-103 L		
Transport a moderate amount of cargo (1000s of kg) from Earth to south pole region sites on the lunar surface	FN-T-202 L		
Provide power for deployed surface utilization payload(s) and/or equipment	FN-P-401 L		
Enable crew lunar surface extravehicular activity in sunlit areas and non-PSRs	FN-M-101 L		
Unload a moderate amount of cargo (1000s of kg) on the lunar surface	FN-M-402 L		
Reposition a moderate amount of cargo (1000s of kg) at the south pole region on the lunar surface	FN-M-503 L		
Provide communications and data exchange between the lunar surface and Earth	FN-C-101 L		
Produce scalable quantities of oxygen from lunar regolith	FN-A-102 L		
Store oxygen on the lunar surface	FN-A-105 L		
Transport scalable quantities of oxygen produced to exploration elements	FN-A-107 L		
Collect regolith at sub-scale to support demonstration using scalable capability	FN-A-201 L		
Provide storage for collected regolith	FN-A-203 L		
Conduct resource identification utilization payload and/or equipment operations on the lunar surface	FN-I-103 L		

➤ Note: Decomposition currently only for Human Lunar Return (HLR) and Foundational Exploration (FE) Segments

Reexamine STMD Approach to Developing Technologies & Systems (1 of 2)



STMD Focus on Ensuring American Global Leadership in Space Technology centered around **Technology Readiness Level (TRL)** based Solicitations



Advance U.S. space technology innovation and competitiveness in a global context



Encourage technology driven economic growth with an emphasis on the expanding space economy



Inspire and develop a diverse and powerful U.S. aerospace technology community

3



EARLY STAGE INNOVATION AND PARTNERSHIPS

- Early Stage Innovation
 - Space Tech Research Grants
 - Center Innovation Fund
 - Early Career Initiative
 - Prizes, Challenges & Crowdsourcing
 - NASA Innovation Advanced Concepts
- Technology Transfer

SBIR/STTR PROGRAMS

- Small Business Innovation Research
- Small Business Technology Transfer

TECHNOLOGY MATURATION

- Game Changing Development
- Lunar Surface Innovation Initiative

TECHNOLOGY DEMONSTRATION

- Technology Demonstration Missions
- Small Spacecraft Technology
- Flight Opportunities

Technology Drives Exploration

LOW MID HIGH

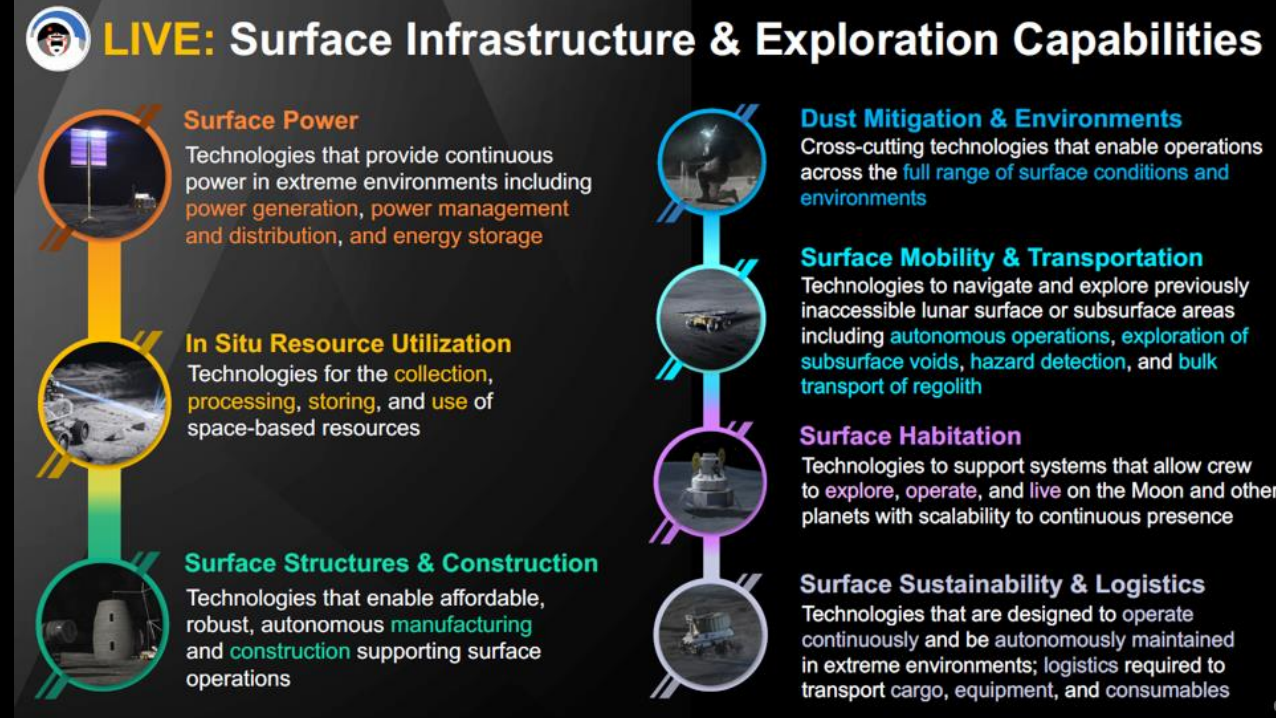
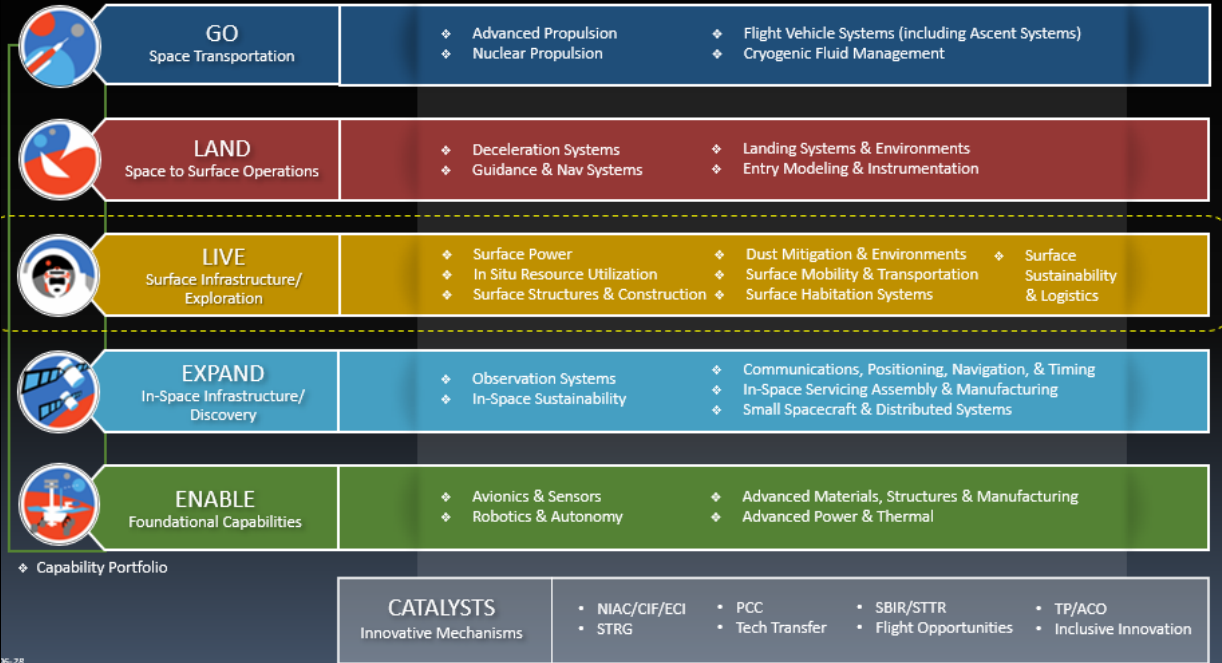
Technology Readiness Level

4

Reexamine STMD Approach to Developing Technologies & Systems (2 of 2)



Focus is now on Technology Base **Functional Domains**

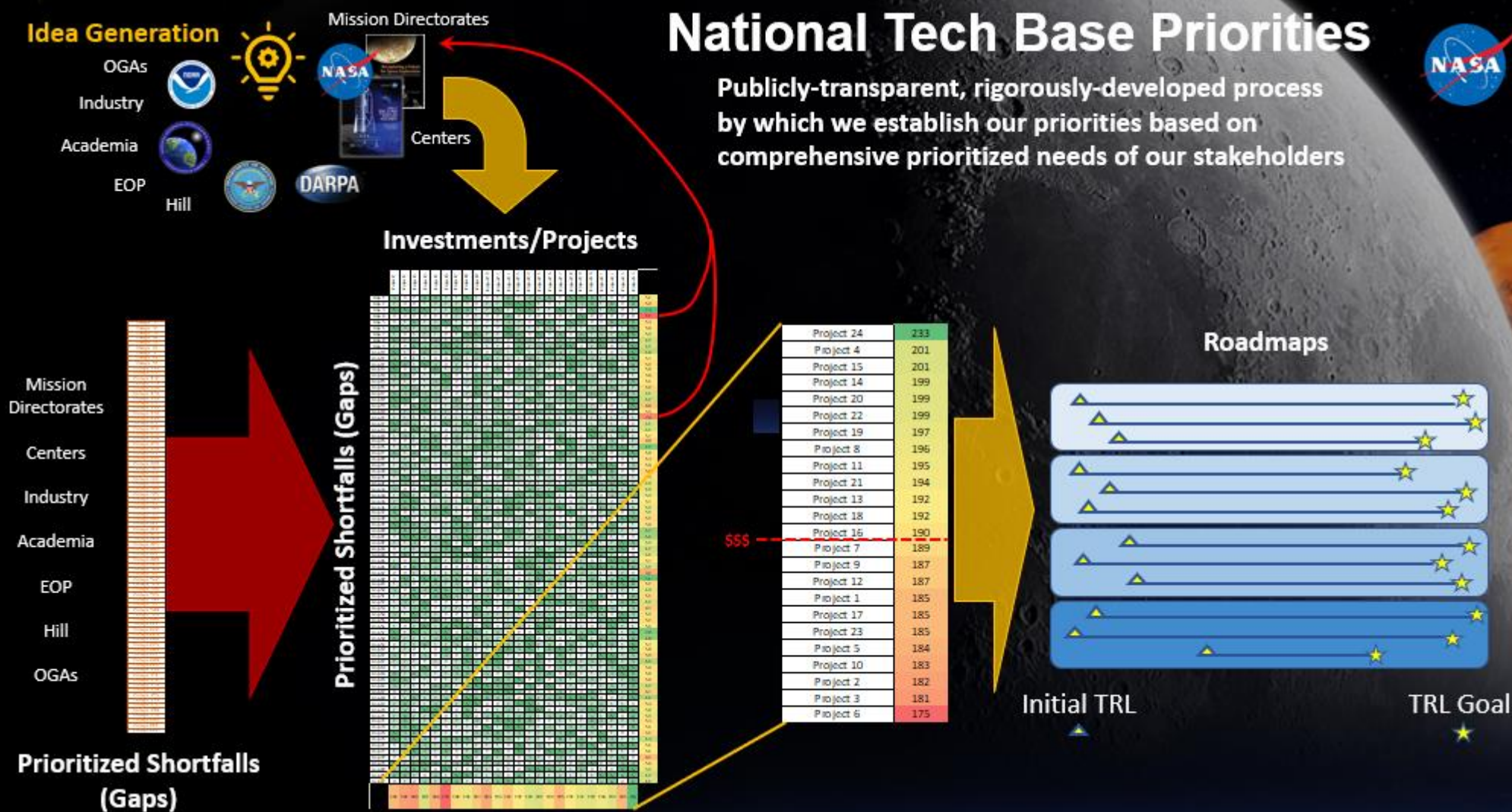


Solicitations will now support more focused **Capability Roadmaps and Priorities**

STMD Instituted a Complete Review of Its Work and Priorities



- Created a Process and tool to prioritize and rank all the work and investments.
- Created 'Shortfalls' for each technical discipline for Stakeholders to review and rank



ISRU Shortfalls, Priorities, and Commodities



Top Priorities seemed to go toward the Most Pressing Needs for Human Lunar Return (HLR) and Foundational Exploration (FE) Artemis Architecture Segments

Integrated Rank	Shortfall ID (blue text if top 3 in category)	Category
1	1618: Survive and operate through the lunar night	Thermal Management Systems
2	1596: High Power Energy Generation on Moon and Mars Surfaces	Power
3	1554: High Performance Onboard Computing to Enable Increasingly Complex Operations	Avionics
4	1557: Position, Navigation, and Timing (PNT) for In-Orbit and Surface Applications	Communication and Navigation
5	1545: Robotic Actuation, Subsystem Components, and System Architectures for Long-Duration and Extreme Environment Operation	Autonomous Systems and Robotics
6	1552: Extreme Environment Avionics	Avionics
7	1519: Environmental Monitoring for Habitation	Advanced Habitation Systems
8	709: Nuclear Electric Propulsion for Human Exploration	Propulsion: Nuclear
9	1304: Robust, High-Progress-Rate, and Long-Distance Autonomous Surface Mobility	Autonomous Systems and Robotics
10	1520: Fire Safety for Habitation	Advanced Habitation Systems
11	1531: Autonomous Guidance and Navigation for Deep Space Missions	Autonomous Systems and Robotics
12	1591: Power Management Systems for Long Duration Lunar and Martian Missions	Power
13	702: Nuclear Thermal Propulsion for Human Exploration	Propulsion: Nuclear
14	1559: Deep Space Autonomous Navigation	Communication and Navigation
15	1527: Radiation Countermeasures (Crew and Habitat)	Advanced Habitation Systems
16	1526: Radiation Monitoring and Modeling (Crew and Habitat)	
17	879: In-space and On-surface, Long-duration Storage of Cryogenic Propellant	
18	1548: Sensing for Autonomous Robotic Operations in Challenging Environmental Conditions	
19	1558: High-Rate Communications Across The Lunar Surface	
20	1626: Advanced Sensor Components: Imaging	
21	792: In-space and On-surface Transfer of Cryogenic Fluids	
22	1569: High-Mass Mars Entry and Descent Systems	
23	1525: Food and Nutrition for Mars and Sustained Lunar	
24	1571: Navigation Sensors for Precision Landing	
25	1573: Terrain Mapping Capabilities for Precision Landing and Hazard Avoidance	
26	1562: Advanced Algorithms and Computing for Precision Landing	
27	1597: Power for Non-Solar-Illuminated Small Systems	
28	1568: Entry Modeling and Simulation for EDL Missions	
29	1516: Water and Dormancy Management for Habitation	
30	1524: Crew Medical Care for Mars and Sustained Lunar	

SPARTACUS Stakeholder 1-187 Rankings - ISRU

Overall Priority	Shortfall ID (blue text if top 3 in category)	Category
53	1578: Extraction and separation of water from extraterrestrial surface material	ISRU
64	1577: Perform resource reconnaissance to location and characterize resources and estimate reserves	ISRU
68	1580: Extraction and separation of oxygen from extraterrestrial minerals	ISRU
79	1583: Produce propellants and mission consumables from extracted in-situ resources	ISRU
103	1585: Extraterrestrial surface environmental simulators, test facilities, and test sites	ISRU
107	1581: Extraction and separation of extraterrestrial atmospheric resources and gaseous products/reactants	ISRU
110	1582: Extraction and separation of metals/metalloids from extraterrestrial minerals	ISRU
117	581: ISRU System Modeling	ISRU
142	1584: Produce manufacturing and construction feedstock from extracted in-situ resources	ISRU
169	1579: Extraction and separation of non-water volatile resources from lunar regolith	ISRU
178	1593: Lunar Surface Power Generation from ISRU Derived Resources	Power
183	1594: Martian Surface Power Generation from ISRU Derived Resources	Power
92	1485: On-Surface Manufacturing of Parts/Products from Surface & Terrestrial Feedstocks	Advanced Manufacturing
145	425: On-surface ISRU-based Construction of Vertical Structures	ECO
160	666: On-surface ISRU-based Construction of Horizontal Structures	ECO
58	1221: Mars Ascent Propulsion	Propulsion: Non-nuclear

ISRU Commodity Priority

- Water – Hydrogen
- Oxygen
- Propellants & Life Support Consumables
- Refined Metals (Al, Fe, Ti)
- Silicon and Ceramics
- Construction Feedstock

Disparity between Stakeholders on ISRU Shortfall Priority



SPARTACUS Stakeholder 1-187 Rankings - ISRU

Overall Priority	Shortfall ID (blue text if top 3 in category)	Academia	Small Industry	Large Industry	OGA	Other	Centers	ESDMD	SMD	Other MDs
53	1578: Extraction and separation of water from extraterrestrial surface material	1	3	20	20	17	8	75		81
64	1577: Perform resource reconnaissance to locate and characterize resources and estimate reserves	40	9	32	105	43	17	67	52	84
68	1580: Extraction and separation of oxygen from extraterrestrial minerals	2	7	69	21	21	14	75		86
79	1583: Produce propellants and mission consumables from extracted in-situ resources	3	5	25	66	29	27	82		126
107	1581: Extraction and separation of extraterrestrial atmospheric resources and gaseous products/reactants	26	107	128	44	69	110	82		148
110	1582: Extraction and separation of metals/metalloids from extraterrestrial minerals	30	55	90	76	101	111	75		159
142	1584: Produce manufacturing and construction feedstock from extracted in-situ resources	11	22	112	60	65	126	105		153
169	1579: Extraction and separation of non-water volatile resources from Lunar regolith	15	43	99	149	123	61	129		131
103	1585: Extraterrestrial surface environmental simulators, test facilities, and test sites	75	56	103	172	135	26	75		96
117	581: ISRU System Modeling	114	61	136	83	90	131	75		145



① Mostly Ranked **Highly** by NASA Centers, Small Industry, Academia, and Other

② **Moderate** Ranking by Large Industry, OGA, and Other

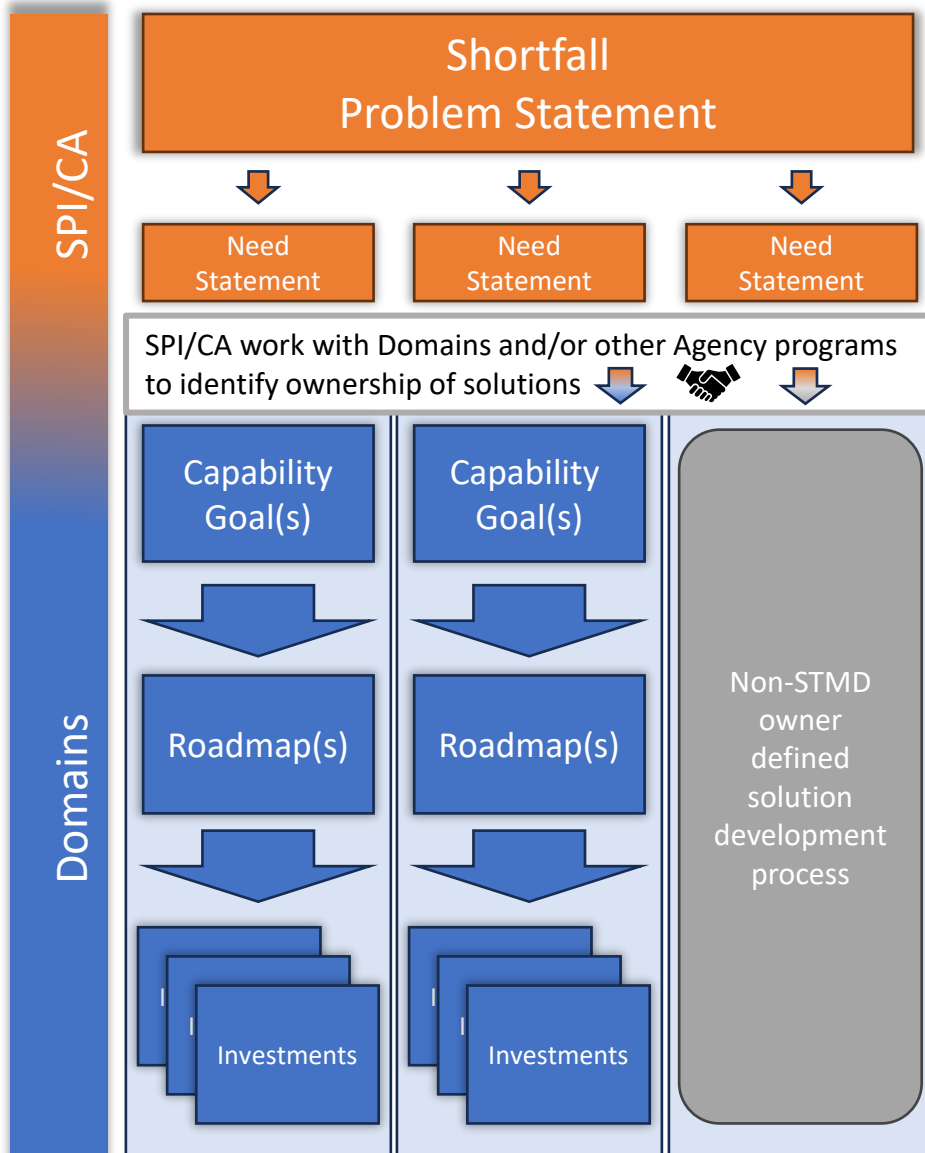
③ **Moderate to Low** Ranking by ESDMD, SMD, and Other MDs

- Ranking Shows Good Alignment with STMD goal of ‘Enabling Industry’ for ISRU Commercialization
- There may be a mismatch in Vision and Near vs Long-term priorities between NASA MDs and External Community. This could significantly impact external investment

Strategy-to-Task Decomposition

187 Shortfalls were too many to properly prioritize:

- Reduced number of Shortfalls to Roadmaps in work



Shortfalls - stakeholder problems

Example: Land large masses safely on the Martian surface.

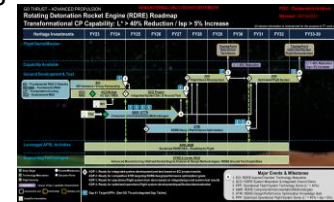
Need Statements - quantified capabilities needed to solve stakeholder problems – divided for ownership

- Enable deceleration of payloads between 25 and 75+ metric tons from Martian orbit to the Martian surface. (STMD, LAND, Deceleration Systems)
- Enable large payloads to land within 100 meters of the intended target while identifying and avoiding obstacles on the order of 1 meter in diameter/depth. (STMD, LAND, Guidance & Navigation Systems)
- Provide safe touchdown of large payloads and crew on the surface of Mars. (STMD, LAND, Landing Systems & Environments)

Capability Goal(s) - quantified solution statement to solve stakeholder problems – may be identical to need statement or broken up into smaller goals that together satisfy the need statement

- Demonstrate and infuse systems to decelerate payloads of 25-75+ t at Mars.
- Demonstrate and infuse systems to precisely (within 100 m) land large payloads on Mars.
- Demonstrate and infuse systems to identify hazards (order of 1 m) on Mars
- Demonstrate and infuse systems to avoid hazards (order of 1 m) on Mars

Roadmap(s) Plan - sequencing and duration of activities to move from the current state to the solution



Investments - development activities that contributes to moving from the current state to the future solution state

Examples: STRG: Computationally-Efficient Large Divert Guidance (Guidance & Nav); VERS-HIAD (Deceleration Systems); Modeling Plume-Surface Interactions (Landing Systems & Environments)

Shared Responsibility between Strategic Planning & Integration (Process Owner) and Domain (Content Owner)

Questions?

